Dutch Disease or Dutch Blessing?

Shale Gas Shock in the United States and its Impact on Educational Decisions

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GitHub Repo: https://github.com/wooyongp/shale-gas-education

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Is the Shale Gas Boom a Blessing or a Curse?

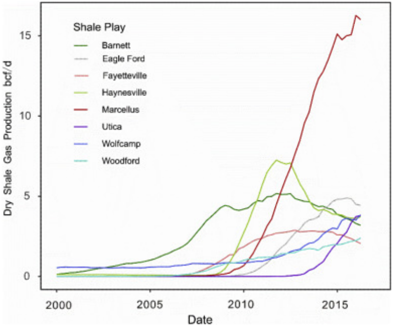

Figure: Shale Gas Production per Area in each Well

Causal Relations

- Shale Gas Boom $\rightarrow$ Income
- Shale Gas Boom $\rightarrow$ Education
- HTE by Income Level

- Rapid growth in natural resource industries generates numerous well-paying jobs that require relatively little formal education.
- As a result, young people may choose to work instead of staying in school leading to lower educational attainment

Literature Review

- Positive Effects of Shale Gas Shock **on Income**
  - Weinstein and Partridge (2011)
- Negative Effects of Shale Gas Shock **on Education**
  - Rickman, Wang, and Winters (2017), Carpenter, Anderson, and Dudensing (2019)

What these papers miss

- No discussions of **heterogeneous treatment effects**, especially by income level
  - Resource booms might increase education for low-income households, but not for high-income households
- They all focus on intra-state district-level comparisons, not cross-state comparisons

IPUMS ACS Data

- IPUMS ACS - 1yr from 2002 to 2010
- Variables of Interest: Household Income, **College Enrollment**, School Attendance, etc.
- Excluded five states: Hawaii, Alaska, Puerto Rico, Colorado, and Wyoming

	Age	Observations	Weighted Population	Personal Income			Family Income		
				Mean	Std. Dev.	Count	Mean	Std. Dev.	Count
Non-shale	[5, 10]	1,308,465.00	170,364,982.00	NaN	-0.00	0.00	72,838.58	75,229.40	1,298,525.00
	[16, 18]	716,276.00	89,493,317.00	2,152.02	5,346.15	716,139.00	76,209.93	75,288.26	684,920.00
	[18, 24]	1,396,758.00	201,384,880.00	10,645.98	13,378.58	1,396,368.00	59,542.08	63,236.34	1,287,939.00
	total	17,029,516.00	2,116,225,634.00	32,089.57	45,426.91	13,756,814.00	71,707.67	72,064.18	16,680,293.00
Shale	[5, 10]	301,477.00	40,178,779.00	NaN	-0.00	0.00	62,547.86	64,017.72	299,038.00
	[16, 18]	162,607.00	20,788,610.00	2,107.18	5,266.69	162,587.00	67,489.58	66,903.25	155,650.00
	[18, 24]	315,031.00	47,152,003.00	10,024.05	12,636.98	314,959.00	52,141.72	55,290.96	291,304.00
	total	3,794,579.00	478,463,294.00	28,501.66	40,124.54	3,043,685.00	63,292.86	62,940.60	3,709,749.00
	Age			Education (Years)			College Enrollment		
				Mean	Std. Dev.	Count	Mean	Std. Dev.	Count
Non-shale	[5, 10]			0.63	1.31	1,308,465.00	0.00	0.00	1,308,465.00
	[16, 18]			10.60	1.32	716,276.00	0.23	0.42	716,276.00
	[18, 24]			12.37	2.02	1,396,758.00	0.82	0.39	1,396,758.00
	total			11.14	5.28	16,429,863.00	0.64	0.48	17,029,516.00
Shale	[5, 10]			0.60	1.25	301,477.00	0.00	0.00	301,477.00
	[16, 18]			10.48	1.40	162,607.00	0.21	0.41	162,607.00
	[18, 24]			12.22	2.00	315,031.00	0.80	0.40	315,031.00
	total			10.74	5.21	3,656,064.00	0.61	0.49	3,794,579.00

Table: Summary Statistics

Methodology: Sun and Abraham (2021) DiD

In our case of shale gas boom, the treatment assignment is staggered across states.

Year	State
2005	Texas
2006	Arkansas
2006	Oklahoma
2008	Alabama
2008	Louisiana
2008	Pennsylvania
2008	West Virginia

Table: Shale Gas Extraction Beginnings by State

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	t_1	t_2	t_3	t_4	t_5
g_1	O	O	O	O	O
g_2	X	O	O	O	O
g_3	X	X	O	O	O
g_4	X	X	X	O	O
g_5	X	X	X	X	O
g_∞	X	X	X	X	X

Table: Staggered Adoption

Assumptions for Causal identification

- Sun and Abraham (2021) DiD
  - **Parallel Trends**: The treated and control groups would have followed the same trend in the absence of treatment
  - **No Compositional Changes**: The composition of the treated and control groups remains constant over time
  - **Homogeneity of Dynamic Treatment Effects**: The dynamic evolution of treatment effect is the same across different shale states

Main Result: Income

- Household income increased by 2.5% overall in the third year after the shale boom.
- In 1Q, the increase was 18%; in 2Q, it was 5%; in 3-4Q, it was not statistically significant.

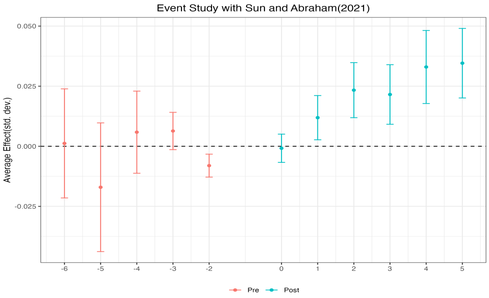

Figure: Overall Income

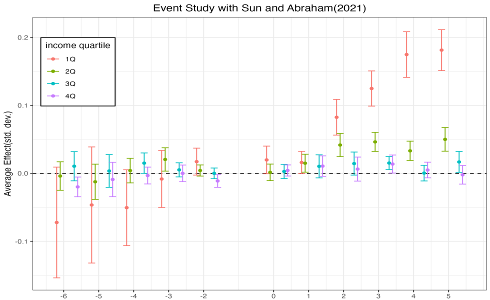

Figure: Income by Quartile

Main Result: Education

- College enrollment increased by 7.5%p overall in the third year after the shale boom.
- The increase was mainly driven by the 1Q households.
- Drawback: Violation in the parallel trend assumptions in the left figure

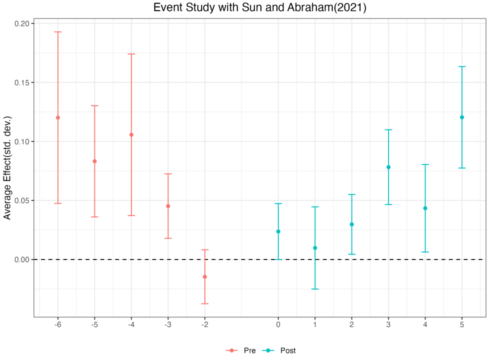

Figure: College Enrollment

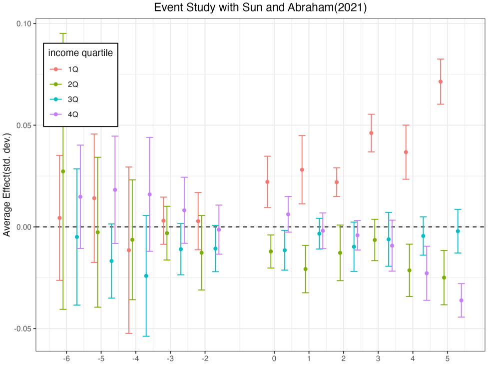

Figure: College Share by Income Quartile

Future Work

- What kind of microdata can I add to this research?
- How can I incorporate ML methods such as causal forest to this research?
  - Wang, Martinez, and Hahn (2024) proposes the following method to estimate CATT in panel data settings:
  - Assumptions: Unconfoundedness, Overlap
  - Outcome Model

$$Y_{it} = \alpha\mu(X_i, T = t) + \beta_S \nu(X_i, S_{it}, T = t) + \epsilon_{it}$$

    where S is the length to exposure (Train XBART forests separately for μ and ν)

  - CATT:

$$\begin{aligned}\tau_t(X_i, S) &\equiv \mathbb{E}[Y_{it}|X_i, S_{it} = S] - \mathbb{E}[Y_{it}|X_i, S_{it} = 0] \\ &= \beta_S \nu(X_i, S_{it}, T = t) - \beta_0 \nu(X_i, S = 0, T = t)\end{aligned}$$

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- Weinstein, Amanda L and Mark D Partridge (2011). "The economic value of shale natural gas in Ohio". In: *Swank program in rural-urban policy summary and report*.

Extensions: Other Educational Outcomes

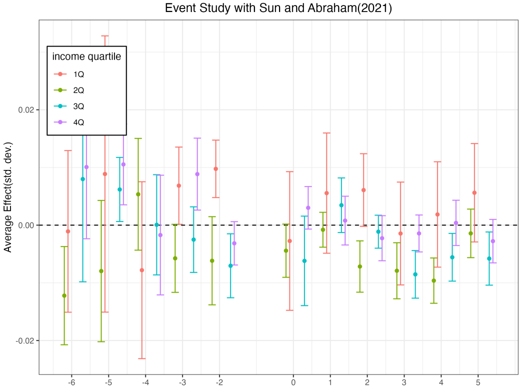

Figure: Elementary School Attendance by Income Quartile

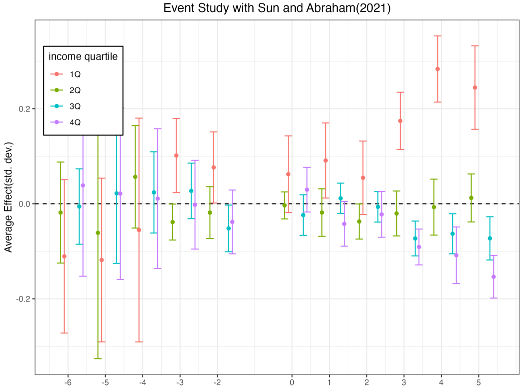

Figure: High School Attendance by Income Quartile